

Strong Chollet inequalities for all simple graph Laplacians

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Abstract

We prove the degree-product form of Chollet's inequality for the Laplacian of every finite simple unweighted graph, including every principal submatrix with degrees retained from the original graph. The result also holds after arbitrary nonnegative diagonal additions. For a two-connected noncycle block, a feasible point of the full matching polytope supplies a permanent lower bound, while a cycle expansion and a trace estimate supply an upper bound for the entrywise square. The two logarithmic coefficients are separated by $1/60$. Cycles are treated directly, and hereditary strength is preserved by diagonal additions and one-point sums. Block decomposition completes the proof for arbitrary graphs. This answers the two all-graph questions stated by Pant and Singh. A separate complete Lean proof establishes the same graph inequalities, including empty and singular cases; we explain its exact scope and its differences from the written proof.

1 Introduction and statements

For an $n \times n$ matrix $A = (a_{ij})$, its permanent is

$$\text{per } A = \sum_{\pi \in \mathfrak{S}_n} \prod_{i=1}^n a_{i,\pi(i)}.$$

Chollet's conjecture asks whether $\text{per}(A \circ B) \leq \text{per}(A) \text{per}(B)$ for Hermitian positive-semidefinite matrices A, B , where $\circ$ is the entrywise product. The self case takes $A = B$. Pant and Singh [1] study structural classes of real matrices and graph Laplacians. Their Section 6 asks whether the self inequality holds for every graph Laplacian and, more strongly, whether its right-hand side can be replaced by the permanent times the product of vertex degrees. We prove this stronger assertion for all finite simple unweighted graphs and for every principal submatrix.

Let $G = (V, E)$ be a finite simple undirected graph. Write $d_v = \deg_G(v)$ and let L_G be its ordinary Laplacian: $(L_G)_{vv} = d_v$, and, for $u \neq v$, $(L_G)_{uv} = -1$ on edges and 0 otherwise. For $S \subseteq V$, $A[S]$ denotes the principal submatrix on S . The degrees in $L_G[S]$ are always the degrees in G , not in $G[S]$. The permanent and diagonal product of an empty matrix are both 1.

Theorem 1.1. *For every G as above and every $S \subseteq V$,*

$$\text{per}(L_G[S] \circ L_G[S]) \leq \text{per}(L_G[S]) \prod_{v \in S} d_v. \quad (1)$$

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More generally, for any numbers $\alpha_v \geq 0$ and $A = L_G + \text{diag}(\alpha_v : v \in V)$,

$$\text{per}(A[S] \circ A[S]) \leq \text{per}(A[S]) \prod_{v \in S} (d_v + \alpha_v). \quad (2)$$

Consequently $\text{per}(L_G \circ L_G) \leq \text{per}(L_G)^2$.

There are no connectivity, minimum-degree, nonsingularity or nonemptiness assumptions. In particular, taking a principal submatrix does not remove the contribution of edges leaving S from its diagonal. No assertion for arbitrary weighted graph Laplacians or arbitrary positive-semidefinite matrices is made.

The proof has three parts. On a two-connected noncycle block, diagonal normalization turns the problem into a comparison between $\text{per}(I + C)$ and $\text{per}(I - B)$, where $C = B \circ B$. A trace estimate gives the upper coefficient $19/12$ over unordered edges, and a fractional matching gives the lower coefficient $8/5$. Their difference is $1/60$. Cycle and edge blocks require separate arguments. Finally, the inequality for every principal submatrix is preserved by one-point sums, allowing the blocks of an arbitrary graph to be assembled.

We use Lieb's established block permanent inequality [2] and Edmonds' full matching-polytope theorem [3], with all their required conditions stated below. The diagonal and coalescence closures are already present in Pant–Singh [1, Theorem 4.3 and Appendix A]; we give the needed algebra in full. The written proof was previously archived as a research note [5]. The present paper organizes that proof and its verification scope; it does not present this editorial revision as a separate solution of the underlying questions.

2 Two established inputs and hereditary strength

A real symmetric positive-semidefinite matrix A is *hereditarily strong* if

$$\text{per}(H \circ H) \leq \text{per}(H) \prod_i h_{ii} \quad \text{for every principal submatrix } H \text{ of } A. \quad (3)$$

The adjective refers to the actual principal diagonal entries.

Theorem 2.1 (Lieb's block permanent inequality). *Let A be Hermitian positive semidefinite and let $S_1, \dots, S_k$ partition its index set. Then*

$$\text{per } A \geq \prod_{j=1}^k \text{per } A[S_j].$$

In particular $\text{per } A \geq \prod_i a_{ii} \geq 0$.

This is the non-strict inequality proved in [2]; no equality classification or unresolved permanental-dominance assertion is used.

Theorem 2.2 (Edmonds' matching polytope). *For a finite simple graph H , a vector $(x_e)_{e \in E(H)}$ is a convex combination of incidence vectors of matchings if and only if*

$$\begin{aligned} x_e &\geq 0, \\ \sum_{e \ni v} x_e &\leq 1 && (v \in V(H)), \\ \sum_{e \in E_H(U)} x_e &\leq \frac{|U| - 1}{2} && (U \subseteq V(H), |U| \text{ odd}). \end{aligned}$$

Here $E_H(U)$ denotes the edges with both ends in U .

In particular, for any real edge weights w_e , some matching F satisfies $\sum_{e \in F} w_e \geq \sum_e x_e w_e$. Indeed this follows by averaging over the convex combination.

3 A cycle-trace permanent upper bound

Lemma 3.1. *Let C be real symmetric, entrywise nonnegative, with zero diagonal and all row sums at most $1/2$. Then*

$$\log \text{per}(I + C) \leq \frac{19}{12} \sum_{i < j} c_{ij}^2. \quad (4)$$

Proof. A permutation is a family of vertex-disjoint nontrivial directed cycles, with its remaining vertices fixed. A two-cycle is counted once; longer cycles are counted up to cyclic rotation with orientations distinguished. Give such a cycle γ weight $w(\gamma)$, the product of its entries of C . Dropping disjointness only adds nonnegative terms, so

$$\text{per}(I + C) \leq \prod_{\gamma} (1 + w(\gamma)) \leq \exp \left(\sum_{\gamma} w(\gamma) \right).$$

For cycles of length k , the total weight is at most $\text{tr}(C^k)/k$: each simple directed cycle has exactly k rooted closed walks in the trace, and the trace also counts nonsimple closed walks with nonnegative weights. This statement also applies to $k = 2$.

All eigenvalues of C lie in $[-1/2, 1/2]$, since its row sums are at most $1/2$. For any such eigenvalue and $k \geq 2$, $\lambda^k \leq 2^{-(k-2)} \lambda^2$. Thus

$$\sum_{\gamma} w(\gamma) \leq \text{tr}(C^2) \sum_{k \geq 2} \frac{2^{-(k-2)}}{k}.$$

The cycle family itself is finite; extending the nonnegative upper bound to this convergent infinite series is harmless. Moreover

$$\sum_{k \geq 2} \frac{2^{-(k-2)}}{k} \leq \frac{1}{2} + \frac{1}{6} + \frac{1}{4} \sum_{k \geq 4} 2^{-(k-2)} = \frac{19}{24}.$$

Since $\text{tr}(C^2) = 2 \sum_{i < j} c_{ij}^2$, this proves (4). Its logarithm is defined because the identity permutation contributes 1 to $\text{per}(I + C)$. $\square$

4 A feasible matching in each noncycle block

Here a graph is two-connected if it has at least three vertices and remains connected after deletion of any single vertex.

Lemma 4.1. *Let G be two-connected and not a cycle. Set $d_v = \deg_G(v)$ and*

$$c_{uv} = \frac{1}{d_u d_v}, \quad x_{uv} = \frac{9}{5} c_{uv} \quad (uv \in E(G)).$$

Then x belongs to the matching polytope of G .

Proof. Every $d_v \geq 2$. Nonnegativity holds, and the vertex constraints follow from

$$\sum_{u \sim v} x_{uv} = \frac{9}{5d_v} \sum_{u \sim v} \frac{1}{d_u} \leq \frac{9}{10} < 1.$$

We check every odd-set constraint, including the small sets.

For $U \subseteq V(G)$ with $|U| \geq 2$, call $v \in U$ exceptional if $d_v \geq 3$ or v has a neighbor outside U . There are at least two exceptional vertices. For proper U , no exception contradicts connectivity, and a unique exception v would disconnect the nonempty sets $U \setminus \{v\}$ and $V(G) \setminus U$ upon deletion of v . For $U = V(G)$, all degrees equal to two would make G a cycle. If only one vertex v had degree at least three, $G - v$ would be a connected graph of maximum degree two with at least three degree-one vertices, namely the neighbors of v . A connected graph of maximum degree two is a path or a cycle and cannot have that property.

Let $d_U(v)$ count neighbors in U . The inequality $2/(d_u d_v) \leq 1/d_u^2 + 1/d_v^2$ implies

$$4 \sum_{uv \in E_G(U)} c_{uv} \leq 2 \sum_{v \in U} \frac{d_U(v)}{d_v^2}.$$

Each deficit $1 - 2d_U(v)/d_v^2$ is nonnegative. It is at least $1/3$ when $d_v \geq 3$, and at least $1/2$ when $d_v = 2$ has an external neighbor. The two exceptions therefore give

$$4 \sum_{uv \in E_G(U)} c_{uv} \leq |U| - \frac{2}{3}. \quad (5)$$

For odd $|U| \geq 5$, it follows that

$$\sum_{uv \in E_G(U)} x_{uv} \leq \frac{9|U| - 6}{20} \leq \frac{|U| - 1}{2},$$

where the last comparison already holds for $|U| \geq 4$.

If $|U| = 3$ and $G[U]$ has at most two edges, its c -sum is at most $1/2$, giving an x -sum at most $9/10$. If $G[U]$ is a triangle, at least two of its vertices have degree at least three. Otherwise the third vertex separates the two degree-two vertices from the rest of G , unless G is that triangle, which was excluded. Hence

$$\sum_{uv \in E_G(U)} c_{uv} \leq \frac{1}{6} + \frac{1}{6} + \frac{1}{9} = \frac{4}{9},$$

and its x -sum is at most $4/5$. A singleton has no internal edge. All conditions of Theorem 2.2 hold. $\square$

Proposition 4.2. *The Laplacian of every two-connected noncycle graph is hereditarily strong. For any principal subset S , put $X_S = \sum_{uv \in E(G[S])} (d_u d_v)^{-2}$. More precisely,*

$$\text{per}(L_G[S] \circ L_G[S]) \leq e^{-X_S/60} \text{per}(L_G[S]) \prod_{v \in S} d_v. \quad (6)$$

Thus the inequality is strict if $G[S]$ contains an edge, and is an equality when $G[S]$ has no edge.

Proof. The empty case is immediate. All original degrees are positive. Normalize

$$M = \text{diag}(d_v^{-1/2} : v \in S) L_G[S] \text{diag}(d_v^{-1/2} : v \in S) = I - B.$$

The matrix M is positive semidefinite. The matrix B is symmetric with zero diagonal and $b_{uv} = 1/\sqrt{d_u d_v}$ on edges of $G[S]$. For $C = B \circ B$, its edge entry is $c_{uv} = 1/(d_u d_v) \leq 1/4$, and its row sums are at most $1/2$. Lemma 3.1 gives

$$\log \text{per}(M \circ M) = \log \text{per}(I + C) \leq \frac{19}{12} X_S. \quad (7)$$

Restrict the feasible vector of Lemma 4.1 to $G[S]$. Restriction preserves all matching-polytope constraints. With edge weights $\log(1 + c_{uv})$, Theorem 2.2 supplies a matching F whose weight is at least $\frac{9}{5} \sum_{uv \in E(G[S])} c_{uv} \log(1 + c_{uv})$. Partition S into its matched pairs and remaining singletons. Each pair has permanent $1 + b_{uv}^2 = 1 + c_{uv}$. Theorem 2.1 yields

$$\text{per } M \geq \prod_{uv \in F} (1 + c_{uv}).$$

For $0 \leq t \leq 1/4$,

$$\log(1 + t) \geq \frac{2t}{2+t} \geq \frac{8}{9}t.$$

The first inequality follows by differentiation: the difference has derivative $t^2/((1+t)(2+t)^2) \geq 0$ and value zero at 0. Consequently

$$\log \text{per } M \geq \frac{8}{5} X_S. \quad (8)$$

The permanent is at least 1 by Theorem 2.1, so these logarithms are well defined. Subtracting (8) from (7) gives $\text{per}(M \circ M) \leq e^{-X_S/60} \text{per } M$. If $Z_S = \prod_{v \in S} d_v$, diagonal scaling gives $\text{per } L_G[S] = Z_S \text{per } M$ and $\text{per}(L_G[S] \circ L_G[S]) = Z_S^2 \text{per}(M \circ M)$. This proves (6). An internal edge makes $X_S > 0$; without any internal edge $M = I$, giving equality. $\square$

5 Cycle and bridge blocks

Lemma 5.1. *A real positive-semidefinite matrix with nonpositive off-diagonal entries and bipartite off-diagonal support is hereditarily strong.*

Proof. Fix a principal submatrix H and a permanent monomial $t_\pi = \prod_i h_{i, \pi(i)}$. Every nonzero nontrivial permutation cycle in its support has even length, so $t_\pi \geq 0$. Positivity of every two-by-two principal minor gives $|h_{ij}| \leq \sqrt{h_{ii} h_{jj}}$. Therefore, with $D = \prod_i h_{ii}$,

$$0 \leq t_\pi \leq \prod_i \sqrt{h_{ii} h_{\pi(i), \pi(i)}} = D.$$

Summing $t_\pi^2 \leq D t_\pi$ proves (3). The argument includes zero diagonal entries and the empty submatrix. $\square$

Lemma 5.2. *The Laplacians of K_2 and every cycle are hereditarily strong.*

Proof. Lemma 5.1 handles K_2 , even cycles and every proper principal submatrix of a cycle Laplacian: the latter has forest support. For the full odd cycle C_n , $n \geq 3$, put $M = L_{C_n}/2 = I - B$, where $b_{uv} = 1/2$ on cycle edges. A maximum matching has $(n-1)/2$ edges, so Theorem 2.1 and the preceding scalar bound give

$$\log \text{per } M \geq \frac{n-1}{2} \log \frac{5}{4} \geq \frac{n-1}{9}.$$

For $C = B \circ B$, the simple directed cycles are the n two-cycles of weight $1/16$ and the two orientations of the full cycle of weight 4^{-n} . The cycle-product argument of Lemma 3.1 gives

$$\log \text{per}(M \circ M) \leq \frac{n}{16} + \frac{2}{4^n}.$$

For every integer $n \geq 3$,

$$\frac{n-1}{9} - \frac{n}{16} - \frac{2}{4^n} = \frac{7n-16}{144} - \frac{2}{4^n} > 0.$$

The expression is increasing in n and equals $1/288$ at $n = 3$. Undoing normalization proves the full-cycle inequality, strictly. $\square$

6 Diagonal additions and one-point sums

The following closure arguments also account for all principal subsets at articulation vertices, where degrees from both sides contribute.

Lemma 6.1. *Hereditary strength is preserved by adding a nonnegative diagonal matrix.*

Proof. It suffices to add $\alpha \geq 0$ at one diagonal index v . For any principal submatrix containing v , write its old matrix as A and put

$$a = a_{vv}, \quad P = \text{per } A, \quad R = \text{per } A(v), \quad Q = \text{per}(A \circ A), \quad T = \text{per}(A(v) \circ A(v)),$$

where $A(v)$ deletes v , and let h be its diagonal product. Expansion in the changed diagonal entry gives

$$\text{per}(A + \alpha E_{vv}) = P + \alpha R, \quad \text{per}((A + \alpha E_{vv}) \circ (A + \alpha E_{vv})) = Q + (2a\alpha + \alpha^2)T.$$

Hereditary strength gives $Q \leq ahP$ and $T \leq hR$, while Theorem 2.1 gives $P \geq aR$. All quantities multiplied below are nonnegative. Hence

$$\begin{aligned} Q + (2a\alpha + \alpha^2)T &\leq h[aP + (2a\alpha + \alpha^2)R] \\ &\leq h(a + \alpha)(P + \alpha R), \end{aligned}$$

because the second difference is $h\alpha(P - aR) \geq 0$. Other principal subsets are unchanged. Iteration proves the lemma. $\square$

Lemma 6.2. *The one-point sum of two hereditarily strong positive-semidefinite matrices is hereditarily strong. Direct sums have the same property.*

Proof. For the one-point sum, identify distinguished indices of A_1, A_2 , add their diagonal entries a_1, a_2 , and set cross entries between the two remaining index sets to zero. The resulting matrix A is positive semidefinite, being the sum of the two embedded matrices. Let B_i delete the distinguished index and put

$$P_i = \text{per } A_i, \quad R_i = \text{per } B_i, \quad Q_i = \text{per}(A_i \circ A_i), \quad T_i = \text{per}(B_i \circ B_i),$$

with h_i the diagonal product of B_i . A permutation cycle through the common vertex cannot visit both sides: such a cycle would have to pass through that vertex twice. Expansion by this cycle, including the fixed-vertex case, gives

$$\text{per } A = P_1 R_2 + R_1 P_2, \tag{9}$$

$$\text{per}(A \circ A) = Q_1 T_2 + T_1 Q_2 + 2a_1 a_2 T_1 T_2. \tag{10}$$

The additional term in (10) comes from $(a_1 + a_2)^2 - a_1^2 - a_2^2$. Hereditary strength and nonnegativity imply

$$\text{per}(A \circ A) \leq h_1 h_2 (a_1 P_1 R_2 + a_2 R_1 P_2 + 2a_1 a_2 R_1 R_2).$$

The target right-hand side is $h_1 h_2 (a_1 + a_2)(P_1 R_2 + R_1 P_2)$. After removing $h_1 h_2$, their difference is

$$a_2 R_2 (P_1 - a_1 R_1) + a_1 R_1 (P_2 - a_2 R_2) \geq 0$$

by Theorem 2.1. This proves the whole-matrix inequality.

For a principal subset containing the common vertex, take its selected indices from each side together with that vertex on each side, and apply the same computation. Either selected side may consist just of the common vertex; the empty-matrix conventions give $R_i = T_i = h_i = 1$ there. For a subset omitting the common vertex, the matrix is a direct sum. Permanents and diagonal products multiply on direct sums, proving both remaining assertions, with empty and singular cases included. $\square$

7 Assembly of all graphs

Proof of Theorem 1.1. The standard block decomposition of a connected graph expresses it as successive one-point sums of its bridges and maximal two-connected blocks; see, for example, [4, Chapter 3]. At each articulation vertex, its full degree is the sum of its degrees in those blocks. Thus the same decomposition identifies the actual Laplacian, with its full diagonal, as their matrix one-point sum. Each bridge is K_2 . Each two-connected block is a cycle or a noncycle. Proposition 4.2 and Lemma 5.2 establish hereditary strength for all blocks, and Lemma 6.2 preserves it during assembly. A one-vertex component has zero Laplacian: its nonempty permanent is zero, and its empty permanent is one. Disconnected graphs are direct sums of their connected components. This proves (1) for all G, S .

Lemma 6.1 proves (2). Finally Theorem 2.1 gives $\text{per } L_G \geq \prod_v d_v \geq 0$. Multiplication by the nonnegative $\text{per } L_G$ yields $\text{per}(L_G \circ L_G) \leq \text{per}(L_G)^2$. $\square$

Remark 7.1. Proposition 4.2 records the quantitative gap already implicit in the block comparison. It classifies equality for those principal submatrices of a two-connected noncycle graph. It is not a classification of equality for all assembled graphs, and the constant $1/60$ is not claimed to be optimal.

8 Formal verification and reproduction

The same graph endpoint has a complete independent Lean proof in the fixed package [6]. Its main declaration is

```
Chollet.strongChollet_all_graphs
```

in `src/CholletAllGraphs.lean`. It uses the actual Mathlib `SimpleGraph.lapMatrix` and `Matrix.permanent`, and the shared `StrongChollet` definition quantifies over every principal vertex subset, retaining the degrees of the original graph. The stronger declaration `laplacianDiagonalStrong_all` permits arbitrary nonnegative diagonal additions to the whole graph Laplacian.

Lean is pinned to 4.34.1, compiler commit 5045d0056413266e57c625dcd7c365b10e377c52, and Mathlib to d13f23b723b8a846827a245b89c10fc7d3f11612. The fixed records report fresh compilation of all 83 source modules and empty-kernel, trust-level-zero replay of 985 owned

declarations and their 44,173-declaration transitive closure. Only the standard axioms `propext`, `Classical.choice` and `Quot.sound` are allowed, and a deliberately false proof is rejected. These are records of the frozen verification run, not a claim that this editorial revision reran the kernel.

The Lean proof is not a line-by-line transcription of this paper. It derives the matching lower bound from proved Gram and Fischer identities, uses the bundled proved perfect-matching polytope theorem, and bounds degree-two vertex-robust blocks through a vanishing third trace and scaled feasible weights. Vertex-count induction strengthened by nonnegative diagonal additions then replaces explicit block-tree assembly. A semantic review checks that the unconditional endpoint has exactly the graph quantifiers in Theorem 1.1. Neither the exponential refinement in Proposition 4.2 nor every intermediate written argument is claimed to be separately formalized. Third-party source adaptations and licenses are documented in the package’s `THIRD_PARTY.md`.

For reproduction, start with the package `README` and `case.json`. The latter fixes all source hashes and the complete module order. Run `lakebuild` for the candidate build. The included `reproduce.py` documents fresh compilation and trust-zero replay with the pinned official dependencies and a new output directory. Its recorded results are `verification/final-review.json`, `verification/replay-summary.json` and `verification/negative-kernel.log`. The traditional proof above does not use a finite computation as a premise.

9 Further questions

The graph theorem leaves the general positive-semidefinite Chollet conjecture untouched. An extension to arbitrary positive edge weights would require additional control: the present fractional matching uses unweighted vertex degrees and the fixed normalized edge size. The proof also does not provide a full equality classification after block assembly. Those questions are distinct from the two all-graph unweighted questions answered here.

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Declaration of generative AI and AI-assisted technologies in the writing process

OpenAI AI assistants assisted drafting, organization and LaTeX editing. The author takes responsibility for the manuscript and its account of the verification evidence.

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